Quantum Mechanics I — Exercise Sheet 3

Mixtures and the Bloch ball

Variant: Questions only

Conventions, as in Lecture 3: an ensemble $\{(p_k, \psi_k)\}$ has $p_k > 0$, $\sum_k p_k = 1$ and unit vectors ψ_k; its density matrix is $\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$; for a qubit $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$ with $r_a = \operatorname{Tr}\rho\sigma_a$; a branch with operator K has probability $\operatorname{Tr}(K\rho K^\dagger)$ and, if this is positive, leaves $K\rho K^\dagger/\operatorname{Tr}(K\rho K^\dagger)$.

Exercise 1 (all the two-member ensembles). Let $\mathbf{r} = (0, 0, 0.6)$. Show that for every point $\mathbf{r}_1$ of the Bloch sphere there is exactly one two-member ensemble of $\mathbf{r}$ that contains the state at $\mathbf{r}_1$: its other member is the second point $\mathbf{r}_2$ where the line from $\mathbf{r}_1$ through $\mathbf{r}$ meets the sphere, and the weight of $\mathbf{r}_1$ is $p = |\mathbf{r}_2 - \mathbf{r}|\,/|\mathbf{r}_2 - \mathbf{r}_1|$. Find the ensemble for $\mathbf{r}_1 = \mathbf{e}_z$ and for $\mathbf{r}_1 = (0.8, 0, 0.6)$.

Exercise 2 (three members). An ensemble has the states $|\uparrow\rangle$, $|{+}x\rangle$ and $|{+}y\rangle$, each with probability $\frac{1}{3}$. (i) Find ρ, its Bloch vector, its eigenvalues and $\operatorname{Tr}\rho^2$. (ii) A measurement along z reads +1. With what probability? Given this reading, with what probability was the system prepared in each of the three states?

Exercise 3 (how mixed can a state be?). Let ρ be the density matrix of an ensemble in $\mathbb{C}^n$. Show that $\operatorname{Tr}\rho^2 \geq 1/n$, with equality only for $\rho = \frac{1}{n}\mathbb{1}$. What does this say for a qubit?

Exercise 4 (a reading nobody looks at). A qubit with density matrix $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$ passes a magnet along a unit vector $\mathbf{m}$, and both outgoing beams are recombined without the reading being recorded. (i) Explain, with Theorem 3.4, why the new density matrix is $E_+\rho E_+ + E_-\rho E_-$. (ii) Show that this is $\frac{1}{2}(\mathbb{1} + (\mathbf{m}\cdot\mathbf{r})\,\mathbf{m}\cdot\boldsymbol{\sigma})$: the Bloch vector becomes $(\mathbf{m}\cdot\mathbf{r})\,\mathbf{m}$. (iii) Apply this with $\mathbf{m} = \mathbf{e}_z$ to the two density matrices of Example 3.7: $\mathbf{r} = (0, 0, 0.6)$ and, after the quarter turn, $\mathbf{r} = (0.6, 0, 0)$.

Exercise 5 (tomography of a mixture). (i) Show that $r_a = 2\,\mathrm{Prob}(+1 \text{ along } \mathbf{e}_a) - 1$ for every qubit density matrix. (ii) Many systems are prepared alike, and the measured fractions of +1 along x, y, z are 0.7, 0.5, 0.5. Find $\mathbf{r}$, ρ and $\operatorname{Tr}\rho^2$. Is the preparation pure?

Exercise 6 ⋆. A qubit in any state ρ is measured along z. If the reading is -1, a resonant π-pulse of Lecture 2 is applied at once, which acts as $-\mathrm{i}\sigma_x$; if it is +1, nothing is done. Write the two branch operators, and show that the density matrix afterwards, with the reading not recorded, is $|\uparrow\rangle\langle\uparrow|$ for every ρ.